MicrobeQuest: A Multimodal Benchmark for Information Extraction in Microbiology

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1 Appendices

A Data Source

This appendix section presents the sources of literature used in the MicrobeQuest datasets. Figure 1 lists the journals of the papers, detailing the journal names, descriptions, the number of papers sourced from each, and the publication date range of the referenced literature. Figure 2 shows the associated microorganism databases, including the database names, descriptions, and the number of papers sourced from each.

B Model Configuration

This appendix section presents the models used in our study, as shown in Figure 3. The figure includes each model's name, type (open-source vs. closed-source), description, development environments, and provider.

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Name	Description	Number of documents	Publication date interval
Applied and Environmental Microbiology	Published by the American Society for Microbiology (ASM), focusing on environmental and applied microbiology research	11	1975-2015
International Journal of Systematic Bacteriology	The predecessor of IJSEM, which focused on research in the field of systemic microbiology, was replaced by IJSEM in 2000	9	1966–1999
Systematic and Applied Microbiology	Published in Microbial Systematics and Applied Research by Elsevier	9	1981–2012
Archives of Microbiology	Springer publishes microbiology journals covering multiple fields such as microbial ecology, genetics, biotechnology, and more	9	1974-1999
Current Microbiology	Springer Publishing covers both basic and applied microbiology research	8	1980–2010
International Journal of Systematic and Evolutionary Microbiology	The internationally authoritative journal for the classification and evolution of microbial systems is the official platform for the publication of new species, published by the Microbiology Society	8	2000-2014
Extremophiles	Publishes research on microorganisms thriving in extreme environments, including thermophiles, halophiles, acidophiles, and psychrophiles.	7	1977-1998
Antonie van Leeuwenhoek	Springer Publishing, named after the "Father of Microbiology," publishes research on microbial system classification, ecology, and biodiversity	6	1980-2011
FEMS Microbiology Letters	The European Microbiology Federation publishes short studies covering a wide range of microbiological fields	3	1982-2008
Frontiers in Microbiology	An open-access journal publishing multidisciplinary research across the full spectrum of microbiological sciences.	3	2010–2020
FEMS Microbiology Ecology	Publish research related to microbial ecology, covering both natural and artificial ecosystems	3	1994–2015
Journal of General and Applied Microbiology	A journal by the Microbiology Society of Japan, covering fundamental and applied research in microbial physiology, metabolism, and biotechnology.	3	1985–2014
Environmental Microbiology	A premier journal covering microbial community dynamics, environmental interactions, biogeochemical cycles, and microbial genomics at ecosystem scale.	3	1999–2018
The Journal of Microbiology	Published by Canadian Science Press, dedicated to research in microbial genetics, physiology, ecology, and more	2	2002-2013
Microbiology and Immunology	Published by the Japanese Society for Bacteriology; covers bacteriology, virology, immunology, host–microbe interactions, and infectious diseases.	2	1980–2012
The ISME Journal	The flagship journal of the International Society for Microbial Ecology, featuring high-impact studies on microbial communities, evolution, and ecosystem processes.	2	2007–2020
Nature Communications	A multidisciplinary Nature journal publishing significant microbiology research, including microbial genomics, host–pathogen interactions, and microbial ecology.	2	2010–2020
Applied Microbiology and Biotechnology	A Springer journal dedicated to microbial biotechnology, bioprocess engineering, metabolic engineering, and industrial microbiology.	2	1985–2015
Microbes and Environments	Published by the Japanese Society of Microbial Ecology, emphasizing the role of microorganisms in the natural environment	2	1986–2014
Metabolic Engineering	A leading journal publishing advances in synthetic biology, metabolic pathway engineering, bioproduction, and systems biotechnology.	2	1999–2020
Astrobiology	Covers microbial life in extreme environments, origin-of-life research, planetary habitability, and life-detection strategies beyond Earth.	2	2001–2020
Journal of Bioscience and Bioengineering	Published by The Society for Biotechnology, Japan; focuses on biotechnology, bioengineering, microbial processes, and applied biosciences.	2	1998–2018

Figure 1: Overview of Research Journals Included in the MicrobeQuest Benchmark: Journal Names, Descriptions, Paper Counts, and Publication Date Range

Name	Description	Number of documents
Deutsche Sammlung von Mikroorganismen und Zellkulturen (DSMZ)	The German Microbial and Cell Bank is one of the world's leading public strain resource centers	22
American Type Culture Collection (ATCC)	The American Standard Strain Collection is widely used in scientific research and industry	12
National Collection of Industrial Food and Marine Bacteria (NCIMB)	The National Centre for Industrial, Food and Marine Bacteria in the United Kingdom	12

Figure 2: Overview of Microorganism Databases Included in the MicrobeQuest Benchmark: Database Names, Descriptions and Document Counts

Name	Type	Description	Env	Company
Qwen-72B	open source	Qwen-72B is a large-scale language model with 72 billion parameters developed by Alibaba Cloud. It demonstrates strong capabilities in reasoning, multilingual understanding, and instruction following. The model we use is Qwen-72B.	Alibaba Cloud Linux 3.2104 LTS 64-bit, 192 GiB, 300 GiB of space	Alibaba
DeepSeek-VL-2.	open source	DeepSeek-VL is a large language model designed for both visual and textual comprehension, with 7 billion parameters. It excels in multimodal tasks, integrating vision and language understanding to perform complex dialogue and image-related tasks. The model we use is DeepSeek-VL2.	Local configuration deployment of i9 14900KF, 4080s, 32RAM, 2T	Deepseek-AI
THUDM/GLM-4-32B-0414	open source	GLM-4 is a large language model developed by Tsinghua University's THUDM, featuring 32 billion parameters. It is optimized for understanding and generating text in multiple languages, with a focus on high-performance reasoning and complex task execution. The model we use is THUDM/GLM-4-32B-0414.	AutoDL, NVIDIA A800 80GB, 256 GB system RAM CUDA 12.1, PyTorch 2.1 1.5 TB SSD storage	ThuDM
qwen-coder-plus	open source	qwen-coder-plus is a code-centric language model from Alibaba's Qwen series, optimized for programming tasks such as code generation, completion, debugging, and multi-language understanding. It is designed to support developers in complex software engineering workflows. The model we use is qwen-coder-plus.	Alibaba Cloud Linux 3.2104 LTS 64-bit, 192 GiB, 300 GiB of space	Alibaba
DeepSeek-R1	Open source	DeepSeek-R1 is a reasoning-enhanced model designed to excel in mathematical reasoning, multi-step deduction, and structured logical tasks. It provides stable and interpretable chain-of-thought behaviors with strong inference performance.	AutoDL, NVIDIA H100 PCIe (80GB), 256 GB RAM, CUDA 12.2, PyTorch 2.1 , 2 TB SSD storage	Deepseek-AI
Hunyuan-Turbos-Version	closed source	Hunyuan-Turbos-Version is Tencent's commercially optimized model, featuring high-efficiency inference, strong instruction following, and stable enterprise-grade performance across multilingual tasks.	Tencent Hybrid API Interface	Tencent
hunyuan-turbos-latest	closed source	Hunyuan TurboS is a high-speed large language model from Tencent, optimized for low-latency performance in natural language understanding, generation, and code tasks. The model we use is hunyuan-turbos-latest.Using Tencent Hybrid API Interface.	Tencent Hybrid API Interface	Tencent
GPT-4o-mini	closed source	A lightweight vari-ant of GPT-4 optimized for efficient inference and cost-effective deployment. The model we use is GPT-4o-mini	OpenAI API	OpenAI
GPT-5	closed source	GPT-5 is an early iteration of OpenAI's GPT series, focusing on text generation, understanding, and conversational capabilities. It is designed to handle a wide range of tasks, from simple queries to complex reasoning.	OpenAI API	OpenAI
Qwen-Max	closed source	Qwen-Max is a high-performance model from Alibaba's Qwen series, optimized for complex reasoning, long-context understanding, and multilingual tasks across diverse domains. The model we use is Qwen-Max.	AliCloud QWQ API	Alibaba
Gemini-2.5	closed source	Gemini-2.5 is an advanced version of Google's Gemini model family, excelling in multimodal comprehension. It integrates text, code, image, and audio analysis to deliver sophisticated insights and perform tasks across various domains. The model we use is Gemini-2.5-Pro.	Gemini Official API.	Google DeepMind
Kimi-latest	closed source	Kimi-latest is a high-performance language model optimized for handling large context windows, with a focus on long-form text generation, analysis, and detailed reasoning. It supports tasks such as complex problem-solving and in-depth content creation. The model we use is Kimi-latest-128k	Moonshot official API.	Moonshot AI
PyMuPDF4LLM	Open source	PyMuPDF4LLM is a specialized language model designed to enhance document processing and analysis. It integrates advanced natural language understanding with PyMuPDF's document handling capabilities, enabling tasks such as content extraction, summarization, and information retrieval from complex documents. The model we use is PyMuPDF4LLM.	Local configuration deployment of i9 14900KF, 4080s, 32RAM, 2T	pymupdf
MistralOCR	closed source	MistralOCR is an optical character recognition (OCR) model designed to efficiently extract text from scanned documents, images, and PDFs. It combines advanced image processing techniques with natural language understanding to handle complex document structures and deliver accurate text recognition. The model we use is MistralOCR.	Mistral official API	Mistral AI

Figure 3: A Comprehensive Overview of Models: Detailed Insights into Names, Types, Descriptions, Development Environments, and Providers

C Standard Prompt Templates

This appendix section presents the standardized prompt templates used for each task in our benchmark. Each template is designed to elicit specific microbiology-related capabilities from LLMs. In addition to the prompt template, the associated `note` section defines the expected output format for each question within the task. Since `note` instructions are question-specific rather than task-specific, we only provide them for selected tasks where output formats require clarification. All `note` specifications used in our benchmark are documented in the accompanying dataset available on GitHub.

C.1 Structured Information Extraction

C.1.1 Strain Entity Recognition and Normalization. This task evaluates a model's ability to identify microbial strain mentions in scientific texts and normalize them into standardized full strain representations (species + strain designation). The task addresses the challenge of diverse strain naming conventions in literature, where strains may be referenced using abbreviated species names (e.g., "E. coli K-12"), partial designations (e.g., "strain MG1655"), or incomplete identifiers. The normalization component standardizes these varied mentions into complete, consistent formats (e.g., "Escherichia coli K-12") to facilitate downstream processing and cross-reference alignment.

The `note` section specifies the required standardized full strain format (species + strain designation), ensuring that models return outputs in the correct form.

Prompt Template

You are a microbiology expert specializing in the accurate identification and normalization of microbial strain names. You will be given a passage of text. Your task is to analyze the content and extract all microbial strain mentions, then normalize them to their standardized forms. Please respond to the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer enclosed within the <Answer> tags as shown below:
<Answer> your_answer_here </Answer>

C.1.2 Strain Entity Resolution. This task assesses a model's capacity to determine whether different strain references correspond to the same microbial entity, requiring a sophisticated understanding of taxonomic relationships and the evolution of nomenclature.

The `note` section defines the required output format as either `True` or `False`, ensuring that the model returns results in the correct form.

Prompt Template

You are a microbiology expert specializing in the accurate identification and normalization of microbial strain names. Your expertise includes taxonomic classification, strain naming conventions, and resolving strain synonyms across different nomenclature systems. You will be given a passage of text. Your task is to carefully analyze the content and determine whether different descriptions of a strain refer to the same microbial entity. This includes recognizing when different naming formats, abbreviations, or historical nomenclature may refer to identical strains.
Please respond to the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer enclosed within the <Answer> tags as shown below:
<Answer> your_answer_here </Answer>

C.1.3 Strain Taxonomy Extraction. This task aims to extract the taxonomic classification of a microbial strain from scientific text. The goal is to identify the most precise taxonomic information available, including domain, family, genus, etc.

The `note` section defines the expected output as a taxon name (e.g., `Bacteria`) if explicitly mentioned, ensuring that the model returns results in the correct form.

Prompt Template

You are a microbiology expert with specialized knowledge in the accurate identification and reasoning of microbial taxonomic information. Your expertise includes taxonomic classification, phylogenetic relationships, and nomenclatural interpretation.
I will provide you with a passage of text. Your task is to carefully analyze the content and determine the taxonomic classification of the strain, including genus, species, and other relevant ranks if available.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the <Answer> tags as shown below:
<Answer> your_answer_here </Answer>

C.1.4 Strain Physiological Characteristic Extraction. This task evaluates a model's ability to precisely extract key physiological traits of microbial strains from scientific texts, including morphological features and biochemical properties essential for strain identification.

The `note` section defines the expected output format for each attribute. For example, for attributes such as motility, the note specifies the accepted output format—for instance, `1` indicates the presence of motility, while `-1` indicates its absence.

Prompt Template

You are a microbiology expert with specialized knowledge in identifying and extracting microbial physiological traits. Your expertise includes bacterial morphology, biochemical test interpretation, and cellular characteristic analysis across diverse microbial species. You will be given a passage of text. Your task is to analyze the content and extract key physiological characteristics of strains, such as Gram-staining results, motility, oxygen requirements, cell morphology, and strain types. Please respond to the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer enclosed within the <Answer> tags as shown below:
<Answer> your_answer_here </Answer>

C.1.5 Environmental Growth Parameter Extraction. This task focuses on evaluating the model's ability to extract specific environmental factors required for microbial growth (e.g., temperature, pH) from unstructured scientific literature.

The `note` section specifies the required output format for each growth-related attribute. For instance, in the case of pH and temperature, the model is expected to return only the numeric value.

Prompt Template

You are an expert in microbiology with specialized knowledge in the precise determination of microbial growth requirements and environmental tolerances. Your expertise includes understanding how environmental factors affect microbial metabolism, reproduction, and survival. You will be given a passage of text. Your task is to carefully analyze the content and extract key environmental parameters required for optimal strain growth, including temperature ranges, pH tolerance, salinity requirements, and oxygen availability preferences.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the <Answer> tags as shown below:
<Answer> your_answer_here </Answer>

C.1.6 Strain Attribute Semantic Categorization. This task evaluates a model's ability to classify extracted microbial strain attributes into standardized semantic categories, requiring a deep understanding of microbiological terminology.

The `note` section provides a set of predefined taxonomic and environmental categories (e.g., Reactors, Intestinal tracts, Soil, Water/sediments, Symbiosis with host, Geothermal, Subsurface,Plants,Oil and gas field), along with detailed classification guidelines. These instructions ensure that the language model performs within an established semantic framework, rather than generating categories arbitrarily.

Prompt Template

You are an expert in microbiology with specialized knowledge in the precise identification and categorization of strain attributes extracted from text into standardized semantic categories, such as growth environment types.
I will provide you with a passage of text. Your task is to carefully analyze the content and categorize identified strain attributes according to established taxonomic and semantic frameworks.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the <Answer> tags as shown below: <Answer> your_answer_here </Answer>

C.1.7 Strain Culture Medium and Growth Condition Extraction. This task assesses a model's ability to identify and structure detailed information about culture media components and specific cultivation conditions required for successful microbial growth from technical descriptions.

Prompt Template

You are an expert in microbiology with specialized knowledge in the formulation of culture media and the optimization of growth conditions for diverse microbial strains. Your expertise includes media composition, environmental requirements for cultivation, and techniques to promote the growth of fastidious organisms.
You will be given a passage of text. Your task is to analyze the content and extract detailed descriptions of the culture medium components and growth conditions required for the strain, including key ingredients, specific environmental parameters, and procedural steps.
Please answer the following question: `{question}`
Follow the reasoning steps provided below: `{note}`
Provide your final answer within the `<Answer>` tags as shown: `<Answer>` your_answer_here `</Answer>`

C.2 Multimodal Understanding

C.2.1 Table-based Strain Attribute Extraction. *This task evaluates a model's capability to accurately interpret and extract structured information from complex tabular data in microbiological literature, requiring sophisticated pattern recognition and relationship inference.*

Prompt Template

You are an expert in microbiology with specialized knowledge in the precise interpretation and analysis of tabular microbiological data. Your expertise includes understanding complex taxonomic tables, growth characteristic matrices, and comparative strain property charts used in research publications and laboratory reports.
I will provide you with a passage of text. Your task is to carefully analyze the content and identify and extract critical information from tabular data, addressing challenges such as header interpretation, cell content association, data relationship inference, and handling of missing or partial data in microbial strain documentation.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the `<Answer>` tags as shown below: `<Answer>` your_answer_here `</Answer>`

C.2.2 Figure-based Strain Attribute Extraction. *This task tests a model's ability to derive meaningful information from descriptions of graphical representations in microbiology research, including growth curves, metabolic pathways, and microscopy image analyses.*

Prompt Template

You are an expert in microbiology with specialized knowledge in the precise interpretation of graphical data representing microbial properties and behaviors. Your expertise includes analyzing growth curves, metabolic pathway diagrams, microscopy image interpretations, and phylogenetic trees commonly found in microbiological research.
I will provide you with a passage of text. Your task is to carefully analyze the content and identify and extract critical information from graphical representations, including curve trend analysis, data point comparison, legend and axis interpretation, and quantitative result extraction from visual data related to microbial strains.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the `<Answer>` tags as shown below: `<Answer>` your_answer_here `</Answer>`

C.2.3 Multimodal Strain Attribute Reasoning. *This task evaluates a model's capacity to integrate and synthesize information about microbial strains across multiple presentation formats, requiring sophisticated cross-modal verification and complementary information processing.*

Prompt Template

You are an expert in microbiology with specialized knowledge in the precise integration and synthesis of multimodal microbiological information. Your expertise includes correlating textual descriptions with tabular data and graphical evidence to form comprehensive understandings of microbial characteristics, behaviors, and classifications.
I will provide you with a passage of text. Your task is to carefully analyze the content and integrate and reason across information from text, tables, and images, performing cross-modal verification, complementary information synthesis, and complex inferencing based on multi-source data about microbial strains.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the `<Answer>` tags as shown below: `<Answer>` your_answer_here `</Answer>`

C.3 Complex Semantic Reasoning

C.3.1 Multi-Entity Attribute Association. *This task assesses a model's ability to correctly assign attributes to their corresponding microbial entities within complex texts, requiring advanced coreference resolution and entity relationship tracking.*

Prompt Template

You are an expert in microbiology with specialized knowledge in the precise identification and association of attributes with their corresponding microbial entities. Your

expertise includes resolving complex referential relationships in scientific literature, disambiguating between similar strains, and tracking attribute assignments across dense technical descriptions.
I will provide you with a passage of text. Your task is to carefully analyze the content and identify attributes corresponding to multiple entities within long paragraphs and accurately align attributes to their correct subjects, especially when subjects are omitted or ambiguously referenced in microbiological contexts.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the `<Answer>` tags as shown below: `<Answer>` your_answer_here `</Answer>`

C.3.2 Multi-value Priority Resolution. This task evaluates a model's capability to select the most appropriate value when confronted with multiple conflicting measurements for the same microbial property, requiring contextual reasoning about experimental reliability.

Prompt Template

You are an expert in microbiology with specialized knowledge in evaluating and prioritizing conflicting or multiple reported values for microbial properties. Your expertise includes understanding experimental context, methodological reliability, and standardized reporting systems for strain characteristics.
I will provide you with a passage of text. Your task is to carefully analyze the content and select the most contextually appropriate and semantically prioritized value when multiple candidate values are associated with the same property, considering factors such as experimental conditions, measurement methods, and scientific consensus.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the `<Answer>` tags as shown below: `<Answer>` your_answer_here `</Answer>`

C.3.3 Negation and Contrast Relationship Parsing. This task measures a model's ability to accurately interpret complex linguistic structures involving negation and contrastive relationships in microbiological contexts, essential for extracting factually correct information.

Prompt Template

You are an expert in microbiology with specialized knowledge in the precise interpretation of complex linguistic structures describing microbial properties. Your expertise includes analyzing negation patterns, contrastive relationships, and exception clauses in scientific literature to extract accurate factual information.
I will provide you with a passage of text. Your task is to carefully analyze the content and parse and interpret negation and contrastive relationships to accurately capture the intended factual meaning from complex expressions about microbial strains, their properties, and behaviors.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the `<Answer>` tags as shown below: `<Answer>` your_answer_here `</Answer>`

C.3.4 Logical Condition Reasoning. This task evaluates a model's capacity to apply logical frameworks and conditional reasoning to microbiological information, requiring the ability to process complex if-then relationships and dependency chains.

Prompt Template

You are an expert in microbiology with specialized knowledge in the application of logical frameworks to microbiological data interpretation. Your expertise includes understanding complex conditional relationships in experimental designs, metabolic pathways, and growth requirements.
I will provide you with a passage of text. Your task is to carefully analyze the content and infer conclusions based on multiple conditional statements and constraints, including conditional dependencies, "if-then" structures, and contextual logic chains relevant to microbial characteristics and behaviors.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the `<Answer>` tags as shown below: `<Answer>` your_answer_here `</Answer>`

C.3.5 Cross-Paragraph Entity Tracking. This task assesses a model's ability to maintain coherent tracking of microbial entities across multiple paragraphs, requiring sophisticated coreference resolution and information integration across distributed contexts.

Prompt Template

You are an expert in microbiology with specialized knowledge in the coherent integration of distributed information about microbial entities. Your expertise includes maintaining entity consistency across complex research papers, tracking strain references across multiple experimental sections, and resolving co-reference in technical writing.
I will provide you with a passage of text. Your task is to carefully analyze the content and track entities across multiple paragraphs to integrate fragmented information and ensure consistent entity-level understanding of microbial strains and their properties.
Please answer the following question: `{question}`

Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the <Answer> tags as shown below: <Answer> your_answer_here </Answer>

C.3.6 Implicit Conclusion Generation. This task evaluates a model's ability to derive scientifically sound inferences about microbial properties that are not explicitly stated but logically follow from the provided information.

Prompt Template

You are an expert in microbiology with specialized knowledge in inferential reasoning based on incomplete microbiological data. Your expertise includes drawing scientifically sound conclusions from partial evidence, understanding implied relationships in research findings, and extrapolating valid inferences from experimental results.
I will provide you with a passage of text. Your task is to carefully analyze the content and infer logically valid conclusions that are not explicitly stated in the text, by synthesizing contextual clues, conditions, and implied relationships about microbial strains and their properties.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the <Answer> tags as shown below: <Answer> your_answer_here </Answer>

C.3.7 Multi-Instance Comparative Reasoning. This task measures a model's capability to perform comparative analyses across different experimental instances in microbiology, requiring the ability to identify patterns, differences, and meaningful relationships across complex contexts.

Prompt Template

You are an expert in microbiology with specialized knowledge in the comparative analysis of microbial entities across different experimental conditions. Your expertise includes identifying meaningful patterns across multiple experiments, understanding significance in comparative studies, and drawing conclusions from parallel or contrasting results.
I will provide you with a passage of text. Your task is to carefully analyze the content and perform comparative analysis across different experimental instances or groups to derive conclusions based on observed differences, similarities, or relative outcomes in microbial behavior, growth patterns, or metabolic activities.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the <Answer> tags as shown below: <Answer> your_answer_here </Answer>

C.4 Layout Structure and Semantic Region Recognition

C.4.1 Semantic Document Region Extraction. This task evaluates a model's ability to identify and extract structured information from specific regions within scientific literature, requiring understanding of document architecture and semantic organization in microbiological publications.

Prompt Template

You are an expert in microbiology with specialized knowledge in the precise identification and extraction of structured information from scientific literature. Your expertise includes recognizing standardized document components, understanding scientific publication formats, and extracting semantically meaningful sections from research papers.
I will provide you with a passage of text. Your task is to carefully analyze the content and accurately identify and extract key semantic regions within scientific literature—such as titles, author names, institutional affiliations, abstracts, introduction sections, methodology descriptions, results, discussions, figure captions, table headers, acknowledgments, and references—maintaining their hierarchical relationships and contextual significance.
Please answer the following question: `{question}`
Follow the reasoning steps provided below to complete the task: `{note}`
Present your final answer within the <Answer> tags as shown below: <Answer> your_answer_here </Answer>

D Subtask Description

In this appendix, we provide detailed definitions, task objectives, and representative examples for all evaluation tasks introduced in the main paper. Each task is described with sample inputs, expected outputs, and the corresponding evaluation criteria.

D.1 Structured Information Extraction

D.1.1 Strain Entity Recognition and Normalization. This task focuses on identifying microbial strain mentions within scientific texts and normalizing them into standardized full strain representations. The goal is to recognize strain entities that may appear in various formats (abbreviated, partial, or informal references) and convert them into complete, canonical strain identifications following the "species + strain designation" format.

- **Task Input:** A passage of scientific text containing one or more microbial strain mentions in diverse naming formats (e.g., abbreviated species names, partial strain designations, or informal references)
- **Task Output:** Complete, standardized strain entity names in canonical "species + strain designation" format
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers

Input:

Based on the results from phylogenetic, morphological, and protein analyses, we conclude that the novel strain represents a novel species of the genus *Methanocaldococcus*, for which the name *Methanocaldococcus villosus* sp. nov. is proposed (type strain KIN24-T80^{T} = DSM 22612^{T} = JCM 16315^{T}).

Question:

`"What is the full name of a strain (species+strain)?"`

Expected Answer:

<Answer> Methanocaldococcus villosus KIN24-T80 </Answer>

D.1.2 Strain Entity Resolution. This task focuses on determining whether different mentions of strain identifiers within a given context refer to the same microbial entity.

- **Task Input:** A passage of text containing information about a microbial strain entity
- **Task Output:** A binary `True` or `False` answer indicating whether the given strain mentions refer to the same entity
- **Evaluation Metric:** Exact-match accuracy score between predicted and reference answers

Input:

Based on the results from phylogenetic, morphological, and protein analyses, we conclude that the novel strain represents a novel species of the genus *Methanocaldococcus*, for which the name *Methanocaldococcus villosus* sp. nov. is proposed (type strain KIN24-T80^{T} = DSM 22612^{T} = JCM 16315^{T}).

Question:

`"Is strain KIN24-T80 the same strain entity as strain JCM 16315?"`

Expected Answer:

<Answer> True </Answer>

D.1.3 Strain Taxonomy Extraction. This task focuses on extracting the full taxonomic lineage of a given microbial strain, including domain, phylum, class, order, family, genus, and species, as available in the input context.

- **Task Input:** A textual passage that contains taxonomic information about a microbial strain, along with by a specific question
- **Task Output:** The explicit taxonomic classification of the strain corresponding to the question scope (e.g., genus, species)
- **Evaluation Metric:** Exact match accuracy between the predicted and reference answers

Input:

A novel chemolithoautotrophic, hyperthermophilic methanogen was isolated from a submarine hydrothermal system at the Kolbeinsey Ridge, north of Iceland. Based on its 16S rRNA gene sequence, the strain belongs to the order *Methanococcales* within the genus *Methanocaldococcus*, with approximately 95% sequence similarity to *Methanocaldococcus jannaschii* as its closest relative.

Question:

`"What is the genus of the microorganism for the strain KIN24-T80?"`

Expected Answer:

<Answer> Methanocaldococcus </Answer>

D.1.4 Strain Physiological Characteristic Extraction. This task focuses on extracting physiological characteristics of microbial strains from a given passage of text. The focus is on intrinsic properties of the strain, such as Gram reaction, motility, and cell morphology.

- **Task Input:** A passage of text describing physiological traits of a microbial strain, along with a specific question
- **Task Output:** The description or value of the strain's physiological trait based on the question scope
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers

Input:

Cells of the novel organism stained Gram-negative and appeared as regular to irregular cocci possessing more than 50 polar flagella. These cell appendages mediated not only motility but also adherence to abiotic surfaces and the formation of cell–cell contacts.

Question:

`"What is the Gram reaction for the strain KIN24-T80?"`

Expected Answer:

<Answer> Gram-negative </Answer>

D.1.5 Environmental Growth Parameter Extraction. This task focuses on extracting environmental growth parameters of microbial strains from a given passage of text. The emphasis is on growth conditions such as temperature, pH, salinity (NaCl concentration), and required chemical components or elements for growth.

- **Task Input:** A passage of text describing the environmental growth conditions of a microbial strain, along with a specific question
- **Task Output:** The strain's growth parameter or required condition based on the question
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers

Input:

Hence, all further experiments were performed at the optimal growth temperature of 80 C. To ascertain the pH dependence of the organism, the pH of the medium was adjusted with diluted sulphuric acid or sodium hydroxide, as indicated above, without the usage of additional buffers. Growth was observed between pH 5.5 and 7.0, with an optimum at pH 6.5; no growth was detected at or below pH 5.0 or at and above pH 7.5. Different amounts of NaCl were added to the culture medium (MGG medium prepared without NaCl) to determine the optimum growth rate with regard to salt concentration. The minimal requirement for growth was 0.5% (w/v) NaCl, the upper limit was 5.5% (w/v) NaCl, and the optimum was 2.5% (w/v) NaCl. The minimum doubling time for growth of strain KIN24-T80^{T} under optimal conditions was 45 min.

Question:

`"What is the minimum NaCl concentration required for growth of the strain KIN24-T80?"`

Expected Answer:

`<Answer> 0.5% </Answer>`

D.1.6 Strain Attribute Semantic Categorization. This task focuses on identifying and categorizing strain attributes related to their growth environments into standardized semantic categories (e.g., geothermal, marine, terrestrial, halophilic, thermophilic).

- **Task Input:** A passage of text describing the growth environment or habitat of a microbial strain, along with a specific question
- **Task Output:** The categorized description of the strain's typical growth environment
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers

Input:

The results of phenotypic characterization confirm the affiliation of KIN24-T80^{T} to the genus *Methanocaldococcus*. Nevertheless, 16S rRNA gene sequence analysis in combination with the unique whole-cell protein SDS-PAGE pattern proved its distinctiveness from any previously described species. Based on the data presented herein, strain KIN24-T80^{T} represents a novel species, for which the name *Methanocaldococcus villosus* sp. nov. is proposed. Herewith, we describe the first hyperthermophilic *Methanocaldococcus* species isolated from a shallow submarine hydrothermal system.

Question:

`"In what specific habitat or environment does this organism typically grow for the strain KIN24-T80?"`

Expected Answer:

`<Answer>Geothermal</Answer>`

D.1.7 Strain Culture Medium and Growth Condition Extraction. This task focuses on identifying and structuring descriptions of culture medium components and cultivation conditions required for strain growth.

- **Task Input:** A passage of text describing the culture medium and associated question
- **Task Output:** A structured answer to the question about the culture medium composition or growth condition
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers

Input:

"Microorganisms DSMZ. PFENNIG'S MEDIUM I consists of multiple solutions. Solution A contains calcium chloride dihydrate (0.25 g), yeast extract (0.25 g), and distilled water (460.00 ml). For marine or estuarine isolates, add 100.0 g NaCl and increase magnesium sulfate heptahydrate to 15.0 g. Solution B includes sodium sulfide nonahydrate (2.00 g) in 135.00 ml distilled water. Solution C is prepared with sodium bicarbonate (1.50 g) in 50.00 ml water, bubbled with CO_2 and filter sterilized. Solution D contains resazurin (0.1%, 0.5 ml) in 450.00 ml distilled water. Solution E includes ammonium chloride (0.35 g), ammonium acetate (0.25 g), pyruvic acid sodium salt (0.25 g), dextrose (0.25 g), magnesium sulfate heptahydrate (0.50 g), potassium chloride (0.35 g), potassium phosphate monobasic (0.35 g), trace element solution SL-12 B (1.00 ml), and distilled water (25 ml), and is filter sterilized. Solution F consists of vitamin B_{12} (0.01 g) in 100.00 ml distilled water, filter sterilized. The trace element solution SL-12 B includes distilled water (1000.00 ml), Na_2-EDTA (3.00 g), $FeSO_4 \cdot 7H_2O$ (1.10 g), $CoCl_2 \cdot 6H_2O$ (190.00 mg), $MnCl_2 \cdot 2H_2O$ (50.00 mg), $ZnCl_2$ (42.00 mg), $NiCl_2 \cdot 6H_2O$ (24.00 mg), $Na_2MoO_4 \cdot 2H_2O$ (18.00 mg), H_3BO_3 (300.00 mg), and $CuCl_2 \cdot 2H_2O$ (2.00 mg), adjusted to pH 6.0. Solutions D, C, and E are mixed, bubbled with CO_2 in an ice bath under sterile conditions, and 50 ml is added to each bottle of solution A. Before use, add 4 ml solution B and 0.1 ml solution F. The final pH is adjusted to 7.1–7.3 using filter-sterilized 1 M Na_2CO_3. The medium is distributed into sterile, nitrogen-gassed screw-cap tubes. During the first 24 hours, iron precipitates as black flocks; no other sediment should appear. Periodic supplementation with neutralized 3% sodium sulfide solution is required. The sulfide solution is made with $Na_2S \cdot 9H_2O$ (3.00 g) in 100.00 ml distilled water, bubbled with nitrogen, autoclaved, and pH-adjusted to ~7.0 with sterile 2 M H_2SO_4. A yellow color indicates a drop to pH ~8. The solution is stirred continuously to avoid elemental sulfur precipitation, and the final solution should be clear and yellow."

Question:

`"What components are required in the culture medium?"`

Expected Answer:

`<Answer>Calcium chloride, yeast extract, sodium sulfide, sodium bicarbonate, resazurin, ammonium chloride, ammonium acetate, pyruvate, dextrose, magnesium sulfate, potassium chloride, potassium phosphate, vitamin B12, trace elements` (including EDTA, $FeSO_4$, $CoCl_2$, $MnCl_2$, $ZnCl_2$, $NiCl_2$, MoO_4, BO_3, $CuCl_2$)</Answer>

D.2 Multimodal Understanding

D.2.1 Table-based Strain Attribute Extraction. This task aims to extract specific attribute information of microbial strains from tabular data. It addresses challenges such as header interpretation, cell value association, multi-attribute relationship inference, and handling of missing or incomplete information.

- **Task Input:** A table describing multiple strain characteristics (e.g., physiological traits, growth conditions)
- **Task Output:** Structured extraction of target strain attributes in response to a specific question
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers

Input:

Table 1. Comparative characteristics of strain B10^T, '*Candidatus* Aciduliprofundum boonei' and *Methanosphaera stadtmanae*

Taxa: 1, strain B10^T (data from this study); 2, '*Candidatus* Aciduliprofundum boonei' DSM 19572 (Reysenbach *et al.*, 2006); 3, *Methanosphaera stadtmanae* DSM 3091^T (Miller & Wolin, 1985). ND, No data available.

Characteristic	1	2	3
Source	Human faeces	Hydrothermal vent field	Human faeces
Morphology	Cocci, as single cells	Pleiomorphic cocci	Cocci, as pairs or tetrads
Diameter (μm)	0.7–1.0	0.6–1.0	1.0–1.2
Motility	Non-motile	Motile	Non-motile
Optima for growth:			
Temperature (°C)	37	70	37
pH	7.6	4.5	6.5–6.9
NaCl (%, w/v)	1.0	ND	ND
DNA G+C content (mol%)	59.93	39	25.8
Cell-wall ultrastructure	Thin electron-dense layer and thick transparent layer	Thick electron-dense layer	Thick electron-dense layer
Substrates for methane production	H_2 + methanol	None	H_2 + methanol

Question:

"What is the optimal NaCl concentration for the strain B-10^T?"

Expected Answer:

<Answer>1%</Answer>

D.2.2 Figure-based Strain Attribute Extraction. This task aims to extract specific attribute information of microbial strains from figure-based data. It addresses challenges such as visual interpretation, pattern recognition, multi-attribute relationship inference, and handling of incomplete or ambiguous visual information.

- **Task Input:** A figure illustrating various strain characteristics (e.g., physiological traits, growth conditions)
- **Task Output:** Structured extraction of target strain attributes in response to a specific question
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers

Input:

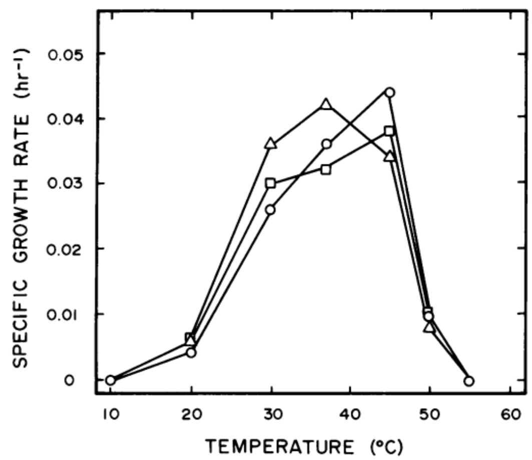


FIG. 2. Relationship between growth rate and incubation temperature for oil field *Methanobacterium* sp. Symbols: ○, strain ivanov; △, strain kuznetsov; □, strain omeliansky.

Question:

"What is the optimal growth temperature for the strain ivanov?"

Expected Answer:

<Answer>46</Answer>

D.2.3 Multimodal Strain Attribute Reasoning. This task focuses on integrating and reasoning across information from text, tables, and images. The information may be present in both the text and the figures, and the model needs to effectively combine multimodal data to answer questions.

- **Task Input:** A passage of text and a table/figure describing the culture medium and associated question.
- **Task Output:** A structured answer to the question about the culture medium composition or growth condition.
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers.

Input:

Text: "An autotrophic thermophilic motile coccoid methanogen was isolated from geothermally heated sea sediments near Naples, Italy. Growth occurs on H_2/CO_2 and on formate between 30 and 70°C, with an optimum at 65°C. The optimal doubling time is only 55 minutes. The NaCl concentration ranges from 1.3% to 8.3%, with an optimum around 4%. By its G + C content of 31.3 mol%, its subunit envelope, and by DNA-RNA hybridization, the new isolate is clearly defined as a member of the genus Methanococcus. We name it *Methanococcus thermolithotrophicus SN-1*."

Figure:

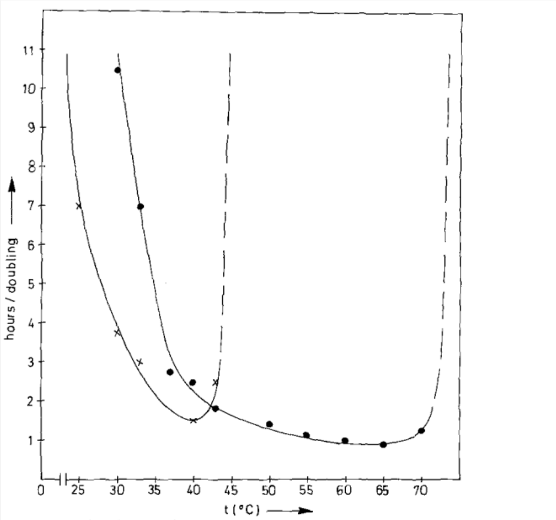


Fig. 1. Optimal growth temperature. ×——× *Methanococcus voltae*; ●——● *Methanococcus thermolithotrophicus*. Growth was determined several times during the exponential phase by O.D.$_{578}$-measurement. The hours/doubling were calculated from the slopes of the growth curves (not shown)

Question:

"What is the lower bound of the optimal growth temperature range for *Methanococcus thermolithotrophicus SN-1*?"

Expected Answer:

`<Answer>65</Answer>`

D.3 Complex Semantic Reasoning

D.3.1 Multi-Entity Attribute Association. This task aims to extract the correct attribute value for a specific microbial entity mentioned in the question from a passage that includes multiple entities and their associated attributes. The model must accurately resolve entity-level ambiguity and ensure that the extracted answer corresponds precisely to the target entity, not to other co-mentioned entities.

- **Task Input:** A passage containing multiple microbial strains, each associated with different attributes, and a question targeting a specific entity's attribute.
- **Task Output:** The exact attribute value (e.g., substrate, temperature, compound name) associated with the target entity mentioned in the question.
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers.

Input:

Our results seem to indicate that, in rice fields, *Methanobacterium* spp. are mostly responsible for CH_4 production from H_2/CO_2, and *Methanosarcina* spp. for CH_4 production from acetate. *Jannaschii* its closest relative.

Question:

`"Is acetate supporting the growth for the strain Methanosarcina spp.?"`

Expected Answer:

`<Answer> true </Answer>`

D.3.2 Multi-value Priority Resolution. This task focuses on selecting the most appropriate attribute value when multiple candidate values are mentioned throughout the document. The model needs to resolve conflicts or prioritize among values based on contextual cues or scientific conventions.

- **Task Input:** A passage containing multiple candidate values for a given attribute
- **Task Output:** The most appropriate and contextually valid attribute value
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers.

Input:

These isolates (strains FDF-17 [T = type strain], FDF-2, SF-2, Ret-1, SD-1, and Cas-1) grew on media containing methanol and mono-, di-, and trimethylamines as catabolic substrates, but not on media containing dimethyl sulfide, methane thiol, H_2, formate, or acetate. Other cultures containing methanol as the catabolic substrate and inoculated in the same way also formed methane, but cultures in media containing acetate, H_2, formate, propionate, butyrate, lactate, or cellulose did not form methane.

Question:

`"What substrates do not support the organism's growth for the strain FDF-17?"`

Expected Answer:

`<Answer>acetate, H₂, formate, propionate, butyrate, lactate, cellulose</Answer>`

D.3.3 Negation and Contrast Relationship Parsing. This task focuses on identifying and interpreting negation and contrast relationships within scientific descriptions. The goal is to accurately determine whether specific conditions, substances, or attributes are positively or negatively associated with the target strain.

- **Task Input:** A scientific passage containing multiple statements about strain behavior under different conditions
- **Task Output:** The exact attribute value or Boolean answer that correctly reflects the negated or contrasted relationship
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers.

Input:

To test the ability of the isolate to utilize energy sources other than H_2, the medium was prepared with a gas phase of N_2/CO_2 (250 kPa, 80:20, v/v), and the following substrates were added separately to final concentrations of 0.1% (w/v): acetate, formate, methanol, pyruvate, and yeast extract. No growth could be detected over a period of 3 days by phase-contrast microscopy. Therefore, the strain was considered to grow exclusively by reduction of CO_2 using H_2 as an electron donor, like all members of the genus *Methanocaldococcus* with validly published names (Jones et al., 1983; Jeanthon et al., 1998, 1999; L'Haridon et al., 2003).

Question:

"Is N_2 supporting the growth for the strain *Methanosarcina* spp.?"

Expected Answer:

`<Answer>false</Answer>`

D.3.4 Logical Condition Reasoning. This task focuses on identifying and interpreting logical conclusions derived from multiple conditional statements and constraints. These may include "if-then" structures, condition-dependent outcomes, and contextual logic chains commonly found in scientific reasoning.

- **Task Input:** A scientific passage containing conditionally structured information or logical dependencies.
- **Task Output:** The exact attribute value or Boolean answer that accurately reflects the inference drawn from the stated conditions.
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers.

Input:

To test whether the organism could grow without hydrogen, cultures were prepared with a gas phase of N_2/CO_2 (80:20) and supplemented separately with acetate, formate, methanol, pyruvate, and yeast extract. No growth was observed over 3 days. Therefore, the strain is considered to grow exclusively via CO_2 reduction using H_2 as the electron donor.

Question:

"What substrates support the organism's growth?"

Expected Answer:

<Answer>H_2/CO_2 </Answer>

D.3.5 Cross-Paragraph Entity Tracking. This task focuses on identifying and interpreting entity attributes that span across multiple paragraphs, requiring the model to track and connect relevant entities and their associated properties across different sections of the text.

- **Task Input:** A passage containing information spread across multiple paragraphs, where entities and their attributes need to be identified and linked correctly.
- **Task Output:** The exact attribute value or Boolean answer that accurately reflects the inference drawn from the connections between the entities and their attributes across the paragraphs.
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers.

Input:

In order to investigate the growth conditions for *Methanocaldococcus* species, MGG medium was prepared with various substrates, including acetate, H_2, and formate. The organism showed significant growth when acetate and H_2 were present, but no growth was observed with formate as the sole substrate. The strains were incubated at a constant temperature of 37°C for a period of 7 days.

Question:

"What substrates support the organism's growth?"

Expected Answer:

<Answer>acetate,H_2</Answer>

D.3.6 Implicit Conclusion Generation. This task focuses on identifying and interpreting implicit conclusions based on the provided scientific context, where conclusions are inferred from the given information rather than explicitly stated.

- **Task Input:** A passage containing information that indirectly leads to a conclusion.
- **Task Output:** The exact attribute value or Boolean answer that accurately reflects the implicit inference drawn from the passage.
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers.

Input:

Selective enrichment culture techniques were employed to obtain mixed cultures of methanogenic rods and sarcina from surface flooding waters and deep subsurface (-1650 m) oil-bearing sedimentary rocks and formation waters sampled from an old oil field in the U.S.S.R. previously reported to display active biological methanogenesis. The methanogens were selectively isolated as colonies on agar petri dishes that were incubated in a novel container.

Question:

"What is the source for the strain?"

Expected Answer:

`<Answer>oil-bearing sedimentary rocks, formation waters</Answer>`

D.3.7 Multi-Instance Comparative Reasoning. This task focuses on conducting a comparative analysis across different experimental conditions or groups to derive conclusions based on observed differences, similarities, or relative outcomes.

- **Task Input:** A scientific passage containing information about different experimental conditions or groups.
- **Task Output:** The exact attribute value or Boolean answer that accurately reflects the comparison between different conditions.
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers.

Input:

As the aforementioned studies have reported a requirement for or stimulus by trace elements or organic compounds for growth of Methanocaldococcus species in the presence of H2 and CO2, MGG medium was prepared without trace minerals and the following substances were added individually or in combination: yeast extract (0.1 g l^{-1}), selenate (0.05 g l^{-1}), tungstate (0.05 g l^{-1}), 1-fold trace mineral solution (Huber & Stetter, 2006; 10 ml l^{-1}) and 1-fold vitamin solution (Balch et al., 1979; 10 ml l^{-1}). Experiments in MGG medium without trace mineral solution resulted in slower growth and two- to fourfold lower final cell densities compared with the original culture medium. Adding selenate to the medium compensated for the effects on growth rate and final cell densities caused by leaving out the trace mineral solution, whereas the addition of tungstate and yeast extract had no influence on the doubling time.

Question:

"Is tungstate an essential growth component?"

Expected Answer:

`<Answer>false</Answer>`

D.4 Layout Structure and Semantic Region Recognition

D.4.1 Semantic Document Region Extraction. This task focuses on identifying and extracting key semantic units from scientific documents, such as author names, article titles, and journal names. The goal is to convert unstructured text into structured information to support document understanding and organization.

- **Task Input:** A passage containing various semantic units such as author names, titles, and other document-related metadata.
- **Task Output:** The exact semantic unit or attribute value extracted from the passage.
- **Evaluation Metric:** Exact-match F1 score between the predicted and reference answers.

Input:

"International Journal of Systematic and Evolutionary Microbiology (2011), 61, 1239–1245 DOI 10.1099/ijs.0.023663-0 Correspondence Annett Bellack annett.bellack@biologie.uni-regensburg.de Methanocaldococcus villosus sp. nov., a heavily flagellated archaeon that adheres to surfaces and forms cell-cell contacts Annett Bellack, Harald Huber, Reinhard Rachel, Gerhard Wanner and Reinhard Wirth Lehrstuhl fuer Mikrobiologie und Archaeenzentrum, Universitaet Regensburg, Universitaetsstrasse 31, 93053 Regensburg, Germany Zentrum fuer Elektronenmikroskopie der NWFIII, Universitaet Regensburg, Universitaetsstrasse 31, 93053 Regensburg, Germany Biozentrum der LMU, Department Biologie I, Großhadernerstrasse 4, 82152 Planegg"

Question:

"What is the name of the journal where this paper was published?"

Expected Answer:

`<Answer>International Journal of Systematic and Evolutionary Microbiology</Answer>`

E JSON Example

This JSON format is specifically designed for MicrobeQuest test cases, aimed at evaluating the performance of models.

case_id: A unique identifier for the test case (e.g., 'MB-S071-00001'),

version: The version number of the test case

timestamp: The timestamp when the test case was created or updated,
difficulty: The difficulty level of the test case (Easy, Medium, Hard),
task_category: The main category of the task,
task_subcategory: The subcategory of the task,
test_case:

- **input:**
  - **pdf_index:** The index of the relevant PDF document (e.g., 'S071'),
  - **image_index:** A list of image indices related to the test case,
  - **is_full_pdf:** A boolean indicating whether the entire PDF is available (e.g., true)
- **question:** The question to be answered in the test case (e.g., 'What is the source of the microorganism for the strain WAL1?'),
- **note:** The instructions for extracting the required information, including step-by-step guidelines,
- **few_shot_examples:** A list of example questions and expected answers to guide the extraction process,
- **expected_answer_type:** Specifies the type of expected answer (e.g., 'descriptive'),
- **expected_answer:** The expected output

F Model Performance

This section provides a brief overview of the model's performance across various benchmark tables and charts, covering accuracy score, F1 score, and BLEU score under different task types. It aims to offer a comprehensive evaluation of the model's strengths, weaknesses, and applicability across multiple metrics.

F.1 Tabular Results

We present three benchmark tables summarizing the model's performance in terms of accuracy score (Table 1), F1 score (Table 2), and BLEU score (Table 3).

Task	Subtask	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	M19
Structured Information Extraction	Strain Entity Recognition and Normalization	0.9151	0.9057	0.2453	0.8585	0.8774	0.9245	0.9623	0.9528	0.9245	0.8774	0.9057	0.8868	0.8679	0.8868	0.8774	0.8868	0.9623	0.9057	0.8868
	Strain Entity Resolution	0.9806	0.9806	0.6065	0.9677	0.9677	0.9290	0.9677	0.9613	0.9677	0.9613	0.9226	0.9677	0.9742	0.9806	0.9613	0.9806	0.9677	0.4452	0.9935
	Strain Taxonomy Extraction	0.7831	0.8991	0.4858	0.6591	0.7670	0.7681	0.8866	0.8040	0.8977	0.6080	0.8068	0.8807	0.7415	0.8068	0.7415	0.8040	0.7964	0.6211	0.6506
	Strain Physiological Characteristic Extraction	0.6402	0.6259	0.1951	0.6139	0.6477	0.5628	0.6035	0.6466	0.6709	0.4114	0.6055	0.6245	0.6097	0.6181	0.6129	0.6276	0.6170	0.4694	0.5844
	Environmental Growth Parameter Extraction	0.6652	0.6805	0.2583	0.6377	0.6681	0.6542	0.6695	0.6943	0.7113	0.5834	0.6622	0.6668	0.6037	0.5953	0.6046	0.6050	0.6753	0.5486	0.5157
	Strain Attribute Semantic Categorization	0.7778	0.7520	0.2932	0.7068	0.7444	0.7236	0.7661	0.7895	0.7895	0.4586	0.7143	0.6917	0.8120	0.8045	0.8195	0.7744	0.7692	0.2256	0.7895
	Strain Culture Medium and Growth Condition Extraction	0.4134	0.4185	0.4850	0.4947	0.5015	0.5121	0.5047	0.5274	0.5253	0.6528	0.4823	0.4851	0.4682	0.4773	0.5350	0.5307	0.5119	0.6027	0.1181
Multimodal Understanding	Table-based Strain Attribute Extraction	0.7284	0.7539	0.2126	0.6926	0.7212	0.7276	0.7680	0.7825	0.7923	0.5822	0.7383	0.7555	0.6460	0.6558	0.6435	0.6623	0.7454	0.5932	0.5871
	Figure-based Attribute Extraction	0.5445	0.5582	0.2212	0.5346	0.5308	0.5231	0.5548	0.6135	0.6346	0.3538	0.5481	0.5442	0.5308	0.5423	0.5423	0.5558	0.5841	0.3829	0.4788
	Multimodal Strain Attribute Reasoning	0.6364	0.6492	0.1944	0.5652	0.5882	0.6734	0.6607	0.6701	0.6854	0.5627	0.6624	0.6471	0.5524	0.5652	0.5601	0.5575	0.6581	0.4956	0.4501
Complex Semantic Reasoning	Multi-Entity Attribute Association	0.6196	0.7106	0.2620	0.5634	0.6366	0.5882	0.6687	0.6732	0.7070	0.4676	0.5944	0.6676	0.5887	0.6394	0.5775	0.6225	0.6340	0.4643	0.5268
	Multi-value Priority Resolution	0.7378	0.7664	0.2447	0.7146	0.7573	0.7169	0.7429	0.8117	0.8252	0.6214	0.7650	0.7709	0.6544	0.6913	0.6621	0.6816	0.7660	0.5849	0.6019
	Negation and Contrast Relationship Parsing	0.8637	0.9062	0.1915	0.8213	0.8617	0.9400	0.9529	0.9404	0.9468	0.9106	0.9319	0.9128	0.7957	0.7745	0.7745	0.7787	0.9144	0.7864	0.7809
	Logical Condition Reasoning	0.7573	0.7854	0.3614	0.7839	0.8222	0.8480	0.8333	0.8260	0.8317	0.8910	0.8222	0.8031	0.7782	0.7686	0.7782	0.7667	0.8776	0.6378	0.6864
	Cross-Paragraph Entity Tracking	0.5865	0.5957	0.2224	0.5266	0.5418	0.5775	0.5773	0.6179	0.5989	0.3992	0.5342	0.5399	0.5266	0.5190	0.5266	0.5095	0.5877	0.4146	0.4430
	Implicit Conclusion Generation	0.5782	0.5576	0.2450	0.5143	0.5143	0.5500	0.5567	0.5938	0.6049	0.4084	0.4923	0.4834	0.4790	0.4901	0.4724	0.4812	0.5648	0.4442	0.4415
	Multi-Instance Comparative Reasoning	0.7137	0.7345	0.2531	0.7062	0.7097	0.6674	0.7164	0.7274	0.7522	0.5611	0.7009	0.7097	0.6478	0.6708	0.6549	0.6690	0.7026	0.5000	0.6159
Layout Structure and Semantic Region Recognition	Semantic Document Region Extraction	0.8275	0.5600	0.4300	0.7449	0.5025	0.8075	0.4800	0.7850	0.5900	0.6100	0.8125	0.5125	0.8200	0.5200	0.8150	0.5325	0.8571	0.7870	0.5350
Overall Accuracy Score		0.7094	0.7133	0.3004	0.6726	0.6867	0.7052	0.7151	0.7454	0.7475	0.6067	0.7056	0.6972	0.6720	0.6670	0.6755	0.6681	0.7329	0.5505	0.5937

Table 1: Accuracy performance of all models on the MicrobeQuest multimodal benchmark. Blue text indicates open-source models, orange text signifies closed-source models and red number indicates the best-performances model. Model identifiers:
M1 = PyMuPDF4LLM + Qwen-72B,
M2 = MistralOCR + Qwen-72B,
M3 = DeepSeek-VL2,
M4 = PyMuPDF4LLM + THUDM/GLM-4-32B-0414,
M5 = MistralOCR + THUDM/GLM-4-32B-0414,
M6 = PyMuPDF4LLM + Qwen-Coder-Plus,
M7 = MistralOCR + Qwen-Coder-Plus,
M8 = PyMuPDF4LLM + Deepseek-R1,
M9 = MistralOCR + Deepseek-R1,
M10 = Hunyuan-Turbos-Version,
M11 = PyMuPDF4LLM + Hunyuan-Turbos-Latest,
M12 = MistralOCR + Hunyuan-Turbos-Latest,
M13 = PyMuPDF4LLM + GPT-4o-mini,
M14 = MistralOCR + GPT-4o-mini,
M15 = PyMuPDF4LLM + GPT-5,
M16 = MistralOCR + GPT-5,
M17 = Qwen-Max,
M18 = Gemini-2.5-pro,
M19 = Kimi-latest-128k.

Task	Subtask	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	M19
Structured Information Extraction	Strain Entity Recognition and Normalization	0.9459	0.9428	0.2555	0.8942	0.9134	0.9438	0.9782	0.9719	0.9640	0.9027	0.9418	0.9375	0.9101	0.9385	0.9175	0.9353	0.9751	0.9057	0.9051
	Strain Entity Resolution	0.9806	0.9806	0.6065	0.9677	0.9677	0.9290	0.9677	0.9613	0.9677	0.9613	0.9226	0.9677	0.9742	0.9806	0.9613	0.9806	0.9677	0.4452	0.9935
	Strain Taxonomy Extraction	0.8102	0.9042	0.5220	0.6872	0.7840	0.7892	0.8881	0.8239	0.9020	0.6302	0.8281	0.8849	0.7633	0.8101	0.7628	0.8073	0.8161	0.6275	0.6705
	Strain Physiological Characteristic Extraction	0.7475	0.7438	0.2708	0.7081	0.7465	0.6586	0.7201	0.7462	0.7696	0.5074	0.7148	0.7376	0.7297	0.7375	0.7295	0.7479	0.7383	0.5045	0.6967
	Environmental Growth Parameter Extraction	0.7113	0.7246	0.2834	0.6811	0.7107	0.6910	0.6978	0.7377	0.7516	0.6101	0.7064	0.7184	0.6508	0.6461	0.6512	0.6540	0.7172	0.5768	0.5623
	Strain Attribute Semantic Categorization	0.7778	0.7520	0.3100	0.7118	0.7444	0.7236	0.7661	0.7895	0.7895	0.4674	0.7143	0.6927	0.8120	0.8045	0.8195	0.7744	0.7692	0.2256	0.7895
	Strain Culture Medium and Growth Condition Extraction	0.2179	0.5269	0.2612	0.5228	0.5296	0.5402	0.5334	0.5536	0.5528	0.4241	0.5138	0.5173	0.4987	0.5081	0.5080	0.5111	0.5388	0.6386	0.4912
Multimodal Understanding	Table-based Strain Attribute Extraction	0.7855	0.8060	0.2553	0.7476	0.7690	0.7690	0.8031	0.8298	0.8350	0.6205	0.7929	0.8011	0.7041	0.7049	0.7002	0.7124	0.7965	0.6254	0.6452
	Figure-based Attribute Extraction	0.6145	0.6325	0.2575	0.6028	0.6060	0.5796	0.6222	0.6793	0.7060	0.4019	0.6163	0.6200	0.6000	0.6144	0.6108	0.6266	0.6556	0.4187	0.5520
	Multimodal Strain Attribute Reasoning	0.7292	0.7344	0.2530	0.6626	0.6814	0.7524	0.7383	0.7460	0.7532	0.6252	0.7520	0.7249	0.6413	0.6403	0.6483	0.6379	0.7444	0.5608	0.5534
Complex Semantic Reasoning	Multi-Entity Attribute Association	0.7214	0.7720	0.3616	0.6695	0.7018	0.6562	0.7254	0.7680	0.7700	0.5505	0.6979	0.7371	0.6789	0.7053	0.6702	0.6879	0.7365	0.5109	0.6282
	Multi-value Priority Resolution	0.7929	0.8047	0.2762	0.7613	0.7991	0.7618	0.7687	0.8529	0.8603	0.6558	0.8148	0.8112	0.7122	0.7370	0.7204	0.7240	0.8023	0.6183	0.6470
	Negation and Contrast Relationship Parsing	0.8775	0.9262	0.2119	0.8391	0.8791	0.9549	0.9642	0.9514	0.9548	0.9212	0.9468	0.9312	0.8115	0.7946	0.7901	0.7984	0.9247	0.7982	0.7974
	Logical Condition Reasoning	0.7602	0.7898	0.3725	0.7894	0.8288	0.8524	0.8355	0.8319	0.8358	0.8939	0.8263	0.8076	0.7827	0.7721	0.7833	0.7694	0.8805	0.6469	0.6898
	Cross-Paragraph Entity Tracking	0.6558	0.6460	0.2475	0.5863	0.5938	0.6253	0.6279	0.6716	0.6558	0.4296	0.5962	0.6017	0.5876	0.5796	0.5820	0.5707	0.6326	0.4632	0.5071
	Implicit Conclusion Generation	0.6174	0.5953	0.2542	0.5633	0.5617	0.5951	0.5983	0.6404	0.6541	0.4337	0.5477	0.5335	0.5216	0.5421	0.5181	0.5309	0.6133	0.4836	0.5000
	Multi-Instance Comparative Reasoning	0.7701	0.7918	0.2902	0.7460	0.7595	0.7164	0.7598	0.7756	0.7990	0.6008	0.7537	0.7642	0.7031	0.7259	0.7102	0.7269	0.7506	0.5288	0.6690
Layout Structure and Semantic Region Recognition	Semantic Document Region Extraction	0.9452	0.6975	0.7056	0.9031	0.6522	0.9284	0.6193	0.9107	0.7349	0.8567	0.9275	0.6471	0.9267	0.6474	0.9261	0.6670	0.9571	0.9483	0.8296
Overall F1 Score		0.7478	0.7651	0.3331	0.7247	0.7349	0.7482	0.7563	0.7912	0.7920	0.6385	0.7563	0.7464	0.7227	0.7161	0.7228	0.7146	0.7787	0.5848	0.6737

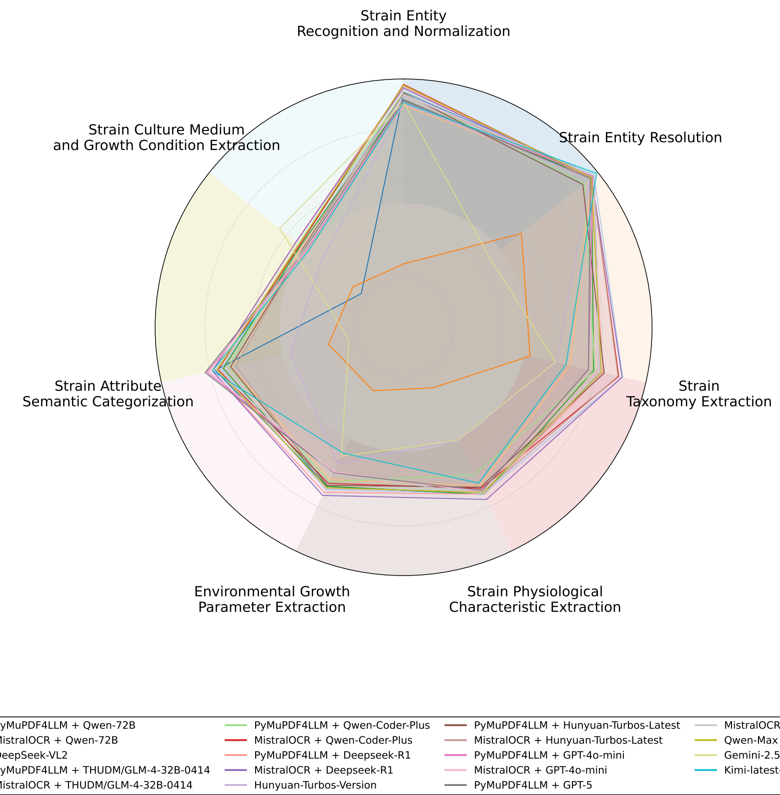
Table 2: F1 performance of all models on the MicrobeQuest multimodal benchmark. Blue text indicates open-source models, orange text signifies closed-source models and red number indicates the best-performances model. Model identifiers:
M1 = PyMuPDF4LLM + Qwen-72B,
M2 = MistralOCR + Qwen-72B,
M3 = DeepSeek-VL2,
M4 = PyMuPDF4LLM + THUDM/GLM-4-32B-0414,
M5 = MistralOCR + THUDM/GLM-4-32B-0414,
M6 = PyMuPDF4LLM + Qwen-Coder-Plus,
M7 = MistralOCR + Qwen-Coder-Plus,
M8 = PyMuPDF4LLM + Deepseek-R1,
M9 = MistralOCR + Deepseek-R1,
M10 = Hunyuan-Turbos-Version,
M11 = PyMuPDF4LLM + Hunyuan-Turbos-Latest,
M12 = MistralOCR + Hunyuan-Turbos-Latest,
M13 = PyMuPDF4LLM + GPT-4o-mini,
M14 = MistralOCR + GPT-4o-mini,
M15 = PyMuPDF4LLM + GPT-5,
M16 = MistralOCR + GPT-5,
M17 = Qwen-Max,
M18 = Gemini-2.5-pro,
M19 = Kimi-latest-128k.

Task	Subtask	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	M19
Structured Information Extraction	Strain Entity Recognition and Normalization	0.3015	0.3145	0.0800	0.2800	0.2960	0.3088	0.3397	0.3237	0.3262	0.2835	0.2989	0.3084	0.2942	0.3109	0.2927	0.3064	0.3328	0.2864	0.2895
	Strain Entity Resolution	0.1744	0.1744	0.1078	0.1721	0.1721	0.1652	0.1721	0.1709	0.1721	0.1709	0.1640	0.1721	0.1732	0.1744	0.1709	0.1744	0.1721	0.0791	0.1767
	Strain Taxonomy Extraction	0.1922	0.2140	0.1193	0.1535	0.1755	0.1931	0.2125	0.2006	0.2177	0.1678	0.1988	0.2116	0.1773	0.1864	0.1770	0.1855	0.1961	0.1641	0.1495
	Strain Physiological Characteristic Extraction	0.2257	0.2236	0.0751	0.2158	0.2283	0.1975	0.2164	0.2161	0.2145	0.1639	0.2181	0.2200	0.2170	0.2153	0.2110	0.2170	0.2238	0.1376	0.2028
	Environmental Growth Parameter Extraction	0.2033	0.2143	0.0824	0.1943	0.2070	0.2045	0.2181	0.2146	0.2274	0.1840	0.2056	0.2135	0.1875	0.1914	0.1868	0.1941	0.2115	0.1785	0.1604
	Strain Attribute Semantic Categorization	0.2181	0.2109	0.0743	0.1964	0.2067	0.2057	0.2160	0.2218	0.2179	0.1332	0.2043	0.1973	0.2308	0.2305	0.2332	0.2231	0.2180	0.0576	0.2228
	Strain Culture Medium and Growth Condition Extraction	0.0986	0.1011	0.1046	0.1137	0.1160	0.1237	0.1232	0.1262	0.1239	0.1420	0.1111	0.1115	0.1061	0.1077	0.1206	0.1172	0.1243	0.1384	0.1023
Multimodal Understanding	Table-based Strain Attribute Extraction	0.2257	0.2364	0.0679	0.2126	0.2237	0.2276	0.2387	0.2378	0.2418	0.1873	0.2300	0.2354	0.2004	0.2029	0.2011	0.2044	0.2316	0.1859	0.1815
	Figure-based Attribute Extraction	0.1548	0.1625	0.0644	0.1542	0.1508	0.1559	0.1659	0.1689	0.1752	0.1185	0.1605	0.1619	0.1501	0.1495	0.1544	0.1513	0.1665	0.1082	0.1370
	Multimodal Strain Attribute Reasoning	0.2257	0.2299	0.0693	0.1980	0.2107	0.2348	0.2296	0.2301	0.2303	0.1949	0.2319	0.2280	0.1938	0.1986	0.1933	0.1967	0.2321	0.1721	0.1699
Complex Semantic Reasoning	Multi-Entity Attribute Association	0.2412	0.2748	0.0932	0.2120	0.2409	0.2330	0.2576	0.2560	0.2701	0.1835	0.2382	0.2671	0.2367	0.2534	0.2346	0.2492	0.2424	0.1813	0.2059
	Multi-value Priority Resolution	0.2246	0.2339	0.0746	0.2179	0.2290	0.2226	0.2309	0.2441	0.2515	0.1971	0.2355	0.2358	0.2025	0.2108	0.2057	0.2089	0.2342	0.1796	0.1805
	Negation and Contrast Relationship Parsing	0.2774	0.2928	0.0638	0.2640	0.2778	0.2996	0.3103	0.3031	0.3009	0.2910	0.2964	0.2923	0.2550	0.2504	0.2484	0.2519	0.2932	0.2506	0.2519
	Logical Condition Reasoning	0.2372	0.2461	0.1153	0.2455	0.2577	0.2672	0.2606	0.2576	0.2590	0.2815	0.2582	0.2518	0.2441	0.2413	0.2441	0.2406	0.2737	0.1943	0.2146
	Cross-Paragraph Entity Tracking	0.1925	0.1920	0.0635	0.1713	0.1776	0.1946	0.1912	0.2048	0.2007	0.1278	0.1769	0.1782	0.1767	0.1747	0.1721	0.1730	0.1863	0.1405	0.1485
	Implicit Conclusion Generation	0.1812	0.1751	0.0686	0.1631	0.1652	0.1765	0.1765	0.1841	0.1877	0.1299	0.1565	0.1550	0.1507	0.1597	0.1492	0.1556	0.1731	0.1435	0.1393
	Multi-Instance Comparative Reasoning	0.2173	0.2249	0.0802	0.2086	0.2176	0.1992	0.2232	0.2167	0.2276	0.1795	0.2187	0.2209	0.1984	0.2067	0.1997	0.2069	0.2154	0.1570	0.1877
Layout Structure and Semantic Region Recognition	Semantic Document Region Extraction	0.6103	0.4930	0.4373	0.5601	0.4562	0.6036	0.4558	0.5940	0.5216	0.5411	0.6049	0.4584	0.6116	0.4657	0.6101	0.4684	0.6158	0.6035	0.4444
Overall BLEU Score		0.2334	0.2341	0.1023	0.2185	0.2227	0.2341	0.2355	0.2428	0.2426	0.2043	0.2338	0.2288	0.2226	0.2183	0.2225	0.2180	0.2413	0.1866	0.1981

Table 3: BLEU performance of all models on the MicrobeQuest multimodal benchmark. Blue text indicates open-source models, orange text signifies closed-source models and red number indicates the best-performances model. Model identifiers:
M1 = PyMuPDF4LLM + Qwen-72B,
M2 = MistralOCR + Qwen-72B,
M3 = DeepSeek-VL2,
M4 = PyMuPDF4LLM + THUDM/GLM-4-32B-0414,
M5 = MistralOCR + THUDM/GLM-4-32B-0414,
M6 = PyMuPDF4LLM + Qwen-Coder-Plus,
M7 = MistralOCR + Qwen-Coder-Plus,
M8 = PyMuPDF4LLM + Deepseek-R1,
M9 = MistralOCR + Deepseek-R1,
M10 = Hunyuan-Turbos-Version,
M11 = PyMuPDF4LLM + Hunyuan-Turbos-Latest,
M12 = MistralOCR + Hunyuan-Turbos-Latest,
M13 = PyMuPDF4LLM + GPT-4o-mini,
M14 = MistralOCR + GPT-4o-mini,
M15 = PyMuPDF4LLM + GPT-5,
M16 = MistralOCR + GPT-5,
M17 = Qwen-Max,
M18 = Gemini-2.5-pro,
M19 = Kimi-latest-128k.

F.2 Radar Chart Analysis

To provide a clearer understanding of the model's performance across different task categories, we present four radar charts 4, 5. Each chart corresponds to one major task type and illustrates the model's performance in terms of F1 score across its sub-tasks.



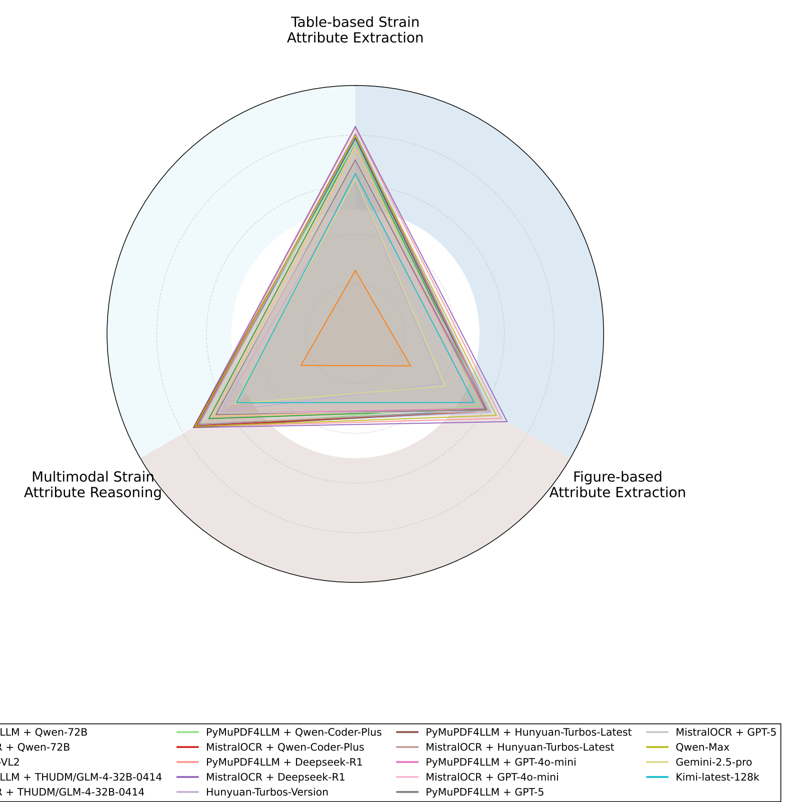


Figure 4: Radar charts showing F1 scores across sub-tasks in Structured Information Extraction and Multimodal Understanding. Each axis represents a task, and each colored line corresponds to a model. The distance from the center indicates the F1-score achieved on that task, with a maximum score of 1.0.

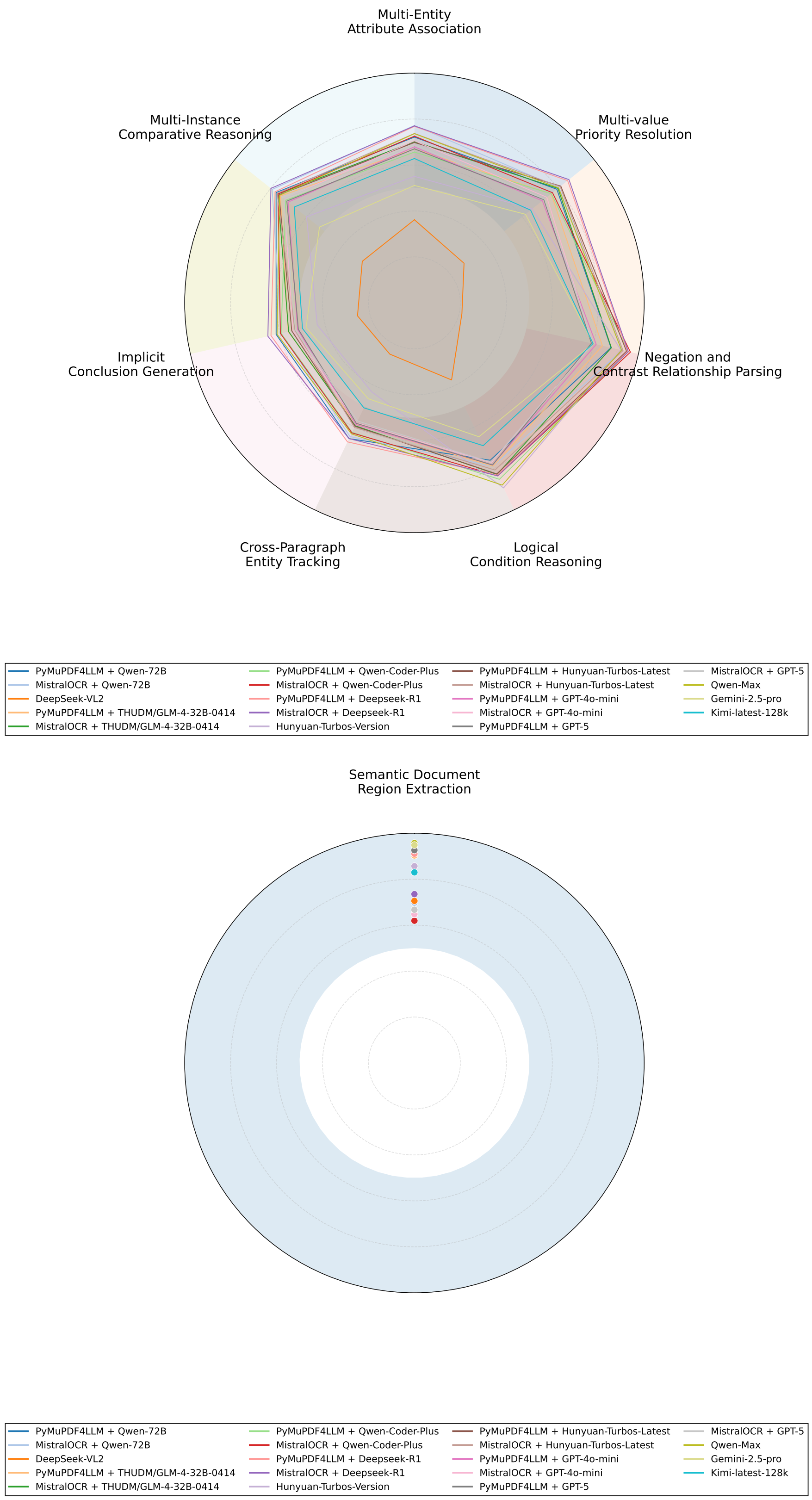


Figure 5: Radar charts showing F1 scores across sub-tasks Complex Semantic Reasoning and Layout and Semantic Region Recognition. Each axis represents a task, and each colored line corresponds to a model. The distance from the center indicates the F1-score achieved on that task, with a maximum score of 1.0.

G Annotation Methodology, Rules, and Quality Control

The *MicrobeQuest* dataset was constructed using a multi-stage, expert-in-the-loop annotation framework designed to ensure scientific validity, annotation consistency, and high-quality evidence grounding. The annotation process integrates structured extraction, systematic consistency checking, and expert adjudication within a unified workflow.

Annotation began with structured extraction performed by undergraduate annotators ($n = 20$) with formal training in microbiology. Annotators extracted predefined microbial attributes from peer-reviewed literature following task definitions and query formulations that were co-designed with senior PhD-level domain experts. This design process ensured biological relevance, terminological consistency, and alignment with real-world microbiological information needs.

To ensure annotation reliability, extracted results subsequently underwent a dedicated consistency-checking phase. During this phase, a subset of undergraduate annotators ($n = 12$) independently verified the correspondence between extracted fields and their supporting textual evidence, focusing on completeness, grounding correctness, and adherence to annotation guidelines. Potential inconsistencies between extracted values and source evidence were flagged by the annotation platform for further review.

Conflicting or ambiguous cases were then resolved by annotators with stronger domain backgrounds in microbiology ($n = 8$). These annotators examined discrepancies highlighted by the system, reconciled divergent interpretations, and standardized annotations according to the established rules. Cases involving insufficient or contradictory evidence were explicitly marked and deferred to expert review rather than resolved through inference.

The final validation stage consisted of independent expert review by two senior microbiologists. Experts examined the biological correctness of all annotations, the validity and specificity of supporting evidence, and the overall consistency of task assignments. Each record was validated by at least two experts, and disputed cases (3.2%) were resolved through structured discussion until expert consensus was achieved.

Throughout the annotation process, annotators adhered to a set of internally defined annotation rules established in collaboration with senior PhD-level domain experts and refined through pilot studies. These rules included:

- **Attribute Definition Rules.** Each microbial attribute was associated with a precise operational definition specifying its biological scope, inclusion and exclusion criteria, and expected value type. Annotators extracted only explicitly stated information that unambiguously matched these definitions and avoided unsupported inference.
- **Evidence Selection Rules.** For each attribute, annotators identified the minimal textual evidence span directly supporting the annotation. Acceptable evidence was limited to experimentally grounded statements, quantitative results, tables, or clearly stated conclusions, excluding speculative or background descriptions.
- **Numerical and Terminological Normalization Rules.** Numerical values were normalized to standardized units according to predefined conversion guidelines. Biological terminology was normalized using controlled vocabularies and canonical naming conventions to reduce lexical variability.
- **Handling Ambiguity and Incomplete Information.** Ambiguous, conflicting, or incomplete information was explicitly flagged rather than inferred. Attributes lacking sufficient evidence were marked as unavailable and deferred to expert judgment.

To further assess annotation consistency, we computed inter-annotator agreement (IAA) on randomly sampled subsets annotated independently prior to expert reconciliation. For categorical annotation dimensions, including task categorization and attribute presence, we employed Fleiss' Kappa:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}, \tag{1}$$

where $\bar{P}$ denotes the observed agreement and $\bar{P}_e$ the expected agreement by chance.

For annotation dimensions involving missing labels or heterogeneous judgment structures, we additionally used Krippendorff's Alpha:

$$\alpha = 1 - \frac{D_o}{D_e}, \tag{2}$$

where D_o and D_e denote observed and expected disagreement, respectively.

These agreement metrics served as diagnostic signals rather than hard acceptance thresholds, supporting expert discussion, identifying systematic ambiguities, and guiding iterative refinement of annotation principles.

Finally, each instance in *MicrobeQuest* was assigned a difficulty label reflecting the complexity of the required reasoning, based on factors such as the number of evidence sources, the presence of multi-step inference, and the degree of multimodal integration. These labels enable fine-grained analysis of model performance across varying task complexities.

H Error Analysis

We categorize the representative error types encountered in our tasks and provide illustrative examples in the following tables.

- **Domain Knowledge Dependency**: Errors that occur when interpreting specialized microbiological terminology or context require specific domain expertise. Respective examples are seen in Table 4.
- **Multi-Value Confusion**: Errors caused by multiple plausible values within the same context, leading to ambiguity in extraction. Respective examples are seen in Table 5.
- **Multi-Entity Confusion**: Errors arising from the presence of multiple, co-referenced, or closely located entities, which complicate precise entity boundary detection. Respective examples are seen in Table 6.
- **Syntactic Complexity**: Errors resulting from complex sentence structures or long-range dependencies that impede accurate parsing and understanding. Respective examples are seen in Table 7.

Question	Context	Expected Answer	Predicted Answer	Error Type
What is the full name of a strain (species + strain)?	Based on the results from phylogenetic, morphological, and protein analyses, we conclude that the novel strain represents a new species within the genus Methanocaldococcus, for which the name Methanocaldococcus villosus sp. nov. is proposed (type strain KIN24-T80 = DSM 22612 = JCM 16315.	Methanocaldococcus villosus KIN24-T80	Methanocaldococcus villosus sp.	Domain knowledge dependency
Is strain Ivanov the same as DSM 2611?	One novel isolate, *Methanobacterium* sp. strain Ivanov, was grown on H_2-CO_2, and the stable-carbon isotopic fractionations that occurred during the synthesis of methane, cell carbon, and lipids were determined. These results were used to examine the anomalous relationship between the isotopic and chemical compositions of natural gas in deep subsurface oil field environments.	true	false	Domain knowledge dependency
What is the class of the microorganism for the strain JAL-1?	*Mycobacterium tuberculosis* is classified within the class Actinomycetia, a group of high G+C Gram-positive bacteria.	Actinomycetia	Methanococci	Domain knowledge dependency

Table 4: Representative Error Cases Involving Domain Knowledge Dependency.

Question	Context	Expected Answer	Predicted Answer	Error Type
In what specific habitat or environment does this organism typically grow for the strain 6A8?	A novel acidophilic, hydrogenotrophic methanogen, designated strain 6A8^T, was isolated from an acidic (pH 4.0–4.5), ombrotrophic (rain-fed) bog near Ithaca, NY, USA. Cultures were dimorphic, containing thin rods (0.2–0.3 μm diameter, 0.8–3.0 μm length) and irregular cocci (0.2–0.8 μm diameter).	Water/sediments	Soil	Multi-Value Confusion
In what environment was the strain M7 isolated?	A chimney sample was collected from the 13 N hydrothermal field during the Hero cruise (1991), on the East Pacific Rise at a depth of 2600 m.; The new strain was isolated from a chimney sample collected from the 13 °N hydrothermal field (12,48 °N, 103,56°W) during the 'Hero' cruise (1991), on the East Pacific Rise at a depth of 2600 m.	Deep-sea hydrothermal chimney sample collected on the East Pacific Rise (13°N; 103°W) at a depth of 2600 m	A chimney sample collected from the 13°N hydrothermal field during the Hero cruise (1991), on the East Pacific Rise at a depth of 2600 m	Multi-Value Confusion
What substrates do not support the organism's growth for the strain M7?	The new isolate utilized methanol in addition to methylamines but not H_2:CO_2, formate, or acetate. The optimal initial pH for growth is 7.7. Trimethylamine, dimethylamine, methylamine, and methanol are substrates for growth and methanogenesis. In contrast, acetate, formate, ethanol, dimethyl sulfide, acetone, 2-butanol, 2-propanol, 1-propanol, and H_2:CO_2 do not support growth. No growth factors are required, but yeast extract and trypticase greatly stimulate growth.	Acetate, formate, ethanol, dimethyl sulfide, acetone, 2-butanol, 2-propanol, 1-propanol, and H_2:CO_2	H_2:CO_2, formate, or acetate	Multi-Value Confusion

Table 5: Representative Error Cases Involving Multi-Value Confusion.

- **Structural Parsing Challenges**: Errors associated with correctly identifying relationships presented in tables, graphs, or other structured document elements. Respective examples are seen in Table 8.

Question	Context	Expected Answer	Predicted Answer	Error Type
What is the Gram reaction for the strain kuznetsov?	Strain kuznetsov stained Gram-negative, whereas both strains Ivanov and Omeliansky stained Gram-positive.	Gram-negative	Gram-positive	Multi-Entity confusion
What is the lower bound of the optimal growth temperature range for the strain AK-7?	Methanogenium boonei (boone.i. N.L. gen. n. boonei of Boone; named in honor of David R. Boone, who has made many contributions to the ecology, physiology, and taxonomy of methanogens). The organism takes the form of irregular cocci 1.0 to 2.5 μm in diameter, occurring singly, and is nonmotile. CO_2 plus H_2 or formate serves as the sole catabolic substrate, with methane as the end product. The fastest growth occurred at 19.4, with a salinity of 0.3 to 0.5 M Na+ and a pH of 6.4 to 7.8. It was isolated from permanently cold, anoxic marine sediments at Skan Bay, Alaska. The type strain is AK-7 (OCM 787/DSMZ 17338).	19.4	15	Multi-Entity confusion
What is the G+C content for the strain AK-3?	The Tm values for strains AK-7, AK-3, and AK-8 were 85.7, 83.2, and 84.2, respectively, which correspond to DNA G+C contents of 49.7, 43.6, and 46.2 mol%, respectively.	43.6	49.7	Multi-Entity confusion

Table 6: Representative Error Cases Involving Multi-Entity Confusion.

Question	Context	Expected Answer	Predicted Answer	Error Type
Which elements or compounds do not stimulate the organism's growth for the strain SD-1?	Strain SD-1 grew in medium containing trimethylamine, dimethylamine, monomethylamine, or methanol as the cat- abolic substrate. No growth occurred in medium supple- mented with 50 mM acetate, 100 kPa of H, plus 20 kPa of CO„ or 5 mM dimethyl sulfide as the catabolic substrate. When 100 kPa of H, plus trimethylamine was added as the catabolic substrate, cultures formed the same quantity of methane and grew at the same rate as controls in medium containing only trimethylamine. With trimethylamine as the catabolic substrate, the cells grew fastest at 42°C (Fig. l), at pH 7.8 (Fig. 2), and in the presence of 0.9 to 3.5 M Na+ Fig.3). Cells grew rapidly (specific growth rate, 0.015 h-) in mineral medium with no organic compound other than trimethylamine added, vitamins did not stimulate growth.	Vitamins	not provided	Syntactic Complexity
How many solutes are there in the culture medium?	YPS MEDIUM: Sea salts (Sigma) 35.00 g, PIPES 3.46 g, Yeast extract 1.00 g, Peptone 4.00 g, Elemental sulphur 5.00 g, NH_4Cl 0.50 g, KH_2PO_4 0.35 g, $CaCl_2$ 0.20 g, $FeCl_3$ 6.70 mg, Na_2WO_4 2.90 mg, Resazurin 0.10 mg, $Na_2S{\cdot}9H_2O$ 0.25 g, plus instructions on pH adjustment, nitrogen flushing, and sterilization.	12	not provided	Syntactic Complexity
Is the strain anaerobic for the strain C?	Strain CT, a non-motile, mesophilic, hydrogenotrophic, methanogenic bacterium, was isolated from an anaerobic digester used for the treatment of raw cassava-peel waste in Congo. The cells were rods, 0.4–0.5 × 2–10 µm in size, and stained Gram-positive. Hydrogen and carbon dioxide were the only substrates that supported growth and methane production. Methane production, but not growth, occurred with CO_2 in the presence of either 2-propanol, 2-butanol or cyclopentanol as hydrogen donors. The temperature range for growth was 25–50 °C, the optimum being between 37 and 42 °C.	true	false	Syntactic Complexity

Table 7: Representative Error Cases Involving Syntactic Complexity.

Question	Context	Expected Answer	Predicted Answer	Error Type
What is the upper bound of the optimal growth temperature range for the strain Sar?	the three rods, as Microorganism Culture condition ARNr 16S phylogeny Optimum (and range) Substrate for growth. Most closely related species (similarity %) Temperature (°C) Salinity (g/l) pH H_2/CO_2, Formate Methanol Acetate Alcohols. Methanobacterium bryantii 37–39 n.d. 6.9–7.2 + - - - +. Methanobacterium formicicum? 37–45 n.d. 6.6–7.8 + + - - +. /RiH2 (Camargue) ○ ○ ○ 2B Mb. bryantii 99.2-FCam (Camargue)* ○ ○ 2B Mb. formicicum 97.9 'FPi (Pila) ○ ○ ○ *Mb. bryantii* 96.5 Methanosarcina barkeri ○ - + + - Methanosarcina mazei? - + + n.d. 'Sar (Camargue)! ○ ○ ○ Ms. barkeri 99.0 'SarPi (Pila) ○ 6 ○ 8 Ms. mazei 99.8 Methanoculleus marisnigri? + - + CoCam ○ ○ Me. marisnigri 98.4 (Camargue)*. *Characteristics of related species according to Garcia [8]. Reference strains. Soil of origin. Concentrations > 60 g/l of NaCl not tested. n.d.: not determined.	37	not provided	Structural Parsing Challenges
What is the optimal growth temperature for the strain omeliansky?	694 BELYAEV ET AL. SPECIFIC GROWTH RATE (hr^{-1}) 10 20 30 40 50 60 TEMPERATURE (°C) FIG. 2. Relationship between growth rate and incubation temperature for oil field Methanobacterium sp. Symbols: O, strain ivanov; A, strain kuznetsov; O, strain omeliansky. Three strains at their respective growth temperature optima was 16 to 18 h. The pH optimum for growth was between 6.5 to 7.2 for strain omeliansky and 7.0 to 7.4 for the other strains. Strain omeliansky grew within the pH range of 6.0 to 7.4, whereas strains ivanov and kuznetsov grew within the pH range of 6.5 to 8.2. Nutritional studies were performed in maintenance medium at the optimum temperature for growth of each strain.	37	not provided	Structural Parsing Challenges
What is the maximum length of Cell morphology for the strain NOBI-1?	ND, No data available. Numbers in parentheses for the optimum temperature, pH and NaCl concentration indicate the range allowing growth. Characteristic 1 2 3 4 5 6 7 8 Cell morphology Rod Rod* Rod Angular, Highly irregular – Irregular Irregular – Sheathed crystal-like plate cocci cocci cocci rod or disc-shaped Cell width (μm) 0.7–1.0 0.2–0.3 0.7 1.5 1.0–3.0 1–2 1–2.5 0.5 Cell length (μm) 2–8+ 0.8–3.0 1.5–2.0 1.6–2.8 1.0–3.0 1–2 1–2.5 7.4–>100 G+C content 56.3 ND 48.84 47.58 51.6 59.4 54.8 45 (mol%) Optimum temperature (°C): 50 (35–55) 37 (10–40) 40 (30–45) 40 (17–41) 20–25 37 (25–55) 37–40 (25–45) 30–37 Optimum pH: 7 (6.7–8.0), 5 (4.0–5.8), 6.1–6.9 (5.9–7.7), 6.5–7.5, 6.8–7.3 (6.0–8.3), 6.7 (5.5–8.0), 7 (6.6–8.8), 6.6–7.4 Optimum NaCl (g/l): 0 (0–15), ND, ND, 10 (4–54), 26.9, <10, 8–12 (0–70), ND Motility: – ND + + – – + + Substrate utilization: Formate – – – – – – – –(2-Propanol/CO_2): – – ND – – + + – Growth requirements: Yeast extract + + + – + – + ND Acetate + + + + + + + ND *Cells sometimes become spherical with a diameter of 0.3–0.8 μm. Determined by buoyant density. Determined by thermal denaturation. Rod-shaped cells with blunt-ends. Often form multicellular filaments. Methanogenic and strictly anaerobic members of the order Methanomicrobiales, phylum Euryarchaeota, domain Archaea. Can use H_2 or formate for growth and methane production. The type species is Methanolinea tarda.	8	not provided	Structural Parsing Challenges

Table 8: Representative Error Cases Involving Structural Parsing Challenges.